

MECH5200 Numerical Methods for PDEs : *Video 14:* *2D Finite Difference Mappings*

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Disclaimer

Quick Disclaimer and comments:

- Please note, typing the equations in detail and writing the code is challenging and there may be errors.
- If you find an error in these notes/this video, please email me to let me know.
- This is a detailed process with lot's of potential places to introduce a coding error, so, please don't get too frustrated as you complete it.
- I have setup this video so that you will see the overall process, but so that you can also complete the code for yourself (for homework and for your own benefit).

Main Steps in the Development of a 2D Poisson solver with Domain Mapping

- **Geometry Definition:** Define the mapping geometry and Equations
- **Equation Discretization and A-Matrix Representation:** Define the discretized system of equations that satisfy the PDE on the computational domain interior nodes.
- **Boundary Conditions:** Define the boundary conditions on the domain
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Geometry Definition: Define the mapping geometry and Equations

- Identify the **physical domain** to be represented.
 - Is it possible to mesh and map this domain?
- Define the **computational domain**:
 - Uniform mesh spacing in each coordinate direction.
 - Commonly a unit line, square, or cube (or higher dimensional space).
- Determine the mapping relationship:
 - **Analytical:** We define a functional mapping (multiple times differentiable) – example from this class (bi-linear mapping):

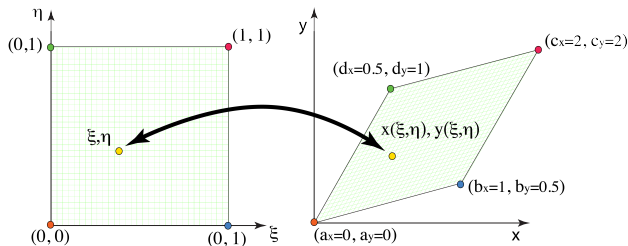
$$\begin{aligned}x(\xi, \eta) &= R_x \xi \eta + S_x \xi + T_x \eta + U_x \\y(\xi, \eta) &= R_y \xi \eta + S_y \xi + T_y \eta + U_y\end{aligned}$$

- **Numerical:** Here, a node in the physical space maps to a specific node in the computational space:

$$\begin{aligned}x(\xi_{i,j}, \eta_{i,j}) &= x_{i,j} \\y(\xi_{i,j}, \eta_{i,j}) &= y_{i,j}\end{aligned}$$

Example

- For our example, we examine the following bilinear (analytical) mapping from the computational to the physical domain:



$$\begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} R_y \\ S_y \\ T_y \\ U_y \end{pmatrix} = \begin{pmatrix} 0.0 \\ 0.5 \\ 2.0 \\ 1.0 \end{pmatrix} \rightarrow \begin{pmatrix} R_y = 0.5 \\ S_y = 0.5 \\ T_y = 1.0 \\ U_y = 0.0 \end{pmatrix}$$

Example

- The equation to map from computational domain to x -coordinate in the physical domain:

$$\begin{aligned}x(\xi, \eta) &= R_x \xi \eta + S_x \xi + T_x \eta + U_x \\x(\xi, \eta) &= 0.5 \cdot \xi \eta + 1.0 \cdot \xi + 0.5 \cdot \eta\end{aligned}$$

- The equation to map from computational domain to y -coordinate in the physical domain:

$$\begin{aligned}y(\xi, \eta) &= R_y \xi \eta + S_y \xi + T_y \eta + U_y \\y(\xi, \eta) &= 0.5 \cdot \xi \eta + 0.5 \cdot \xi + 1.0 \cdot \eta\end{aligned}$$

Example

- The equation to map from computational domain to x - and y -coordinates in the physical domain:

$$x(\xi, \eta) = 0.5 \cdot \xi\eta + 1.0 \cdot \xi + 0.5 \cdot \eta$$

$$y(\xi, \eta) = 0.5 \cdot \xi\eta + 0.5 \cdot \xi + 1.0 \cdot \eta$$

- In this code we will keep things general by maintaining the R, S, T, U variables in the mapping function definition.

Summary: Implementation Tips, Tricks, and Reminders

- We'll want to define the computational domain.
- Not rigid, but, one way to do this:
 - N number of nodes in ξ , and M number of nodes in η ,
 - The grid spacing $\Delta\xi = 1/(N - 1)$, $\Delta\eta = 1/(M - 1)$,
 - Total number of nodes: $NNodes = M * N$.
 - Mapping between node i,j (or n,m) indexing:

$$NN(n, m) = N * (m - 1) + n \quad (1)$$

- While not necessary, it is can be nice to define the computational and physical mesh in a variable structure (matrix?) and plot/visualize.

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Equation Discretization and A-Matrix Representation – Poisson Equation

- When we perform the general mapping for the 2-D Poisson Equation:

$$-\nabla^2 u = - \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) = f$$

- For a general mapping in 2D this can be written as:

$$\frac{-1}{J^2} \left(a \frac{\partial^2 u}{\partial \xi^2} - 2b \frac{\partial^2 u}{\partial \xi \partial \eta} + c \frac{\partial^2 u}{\partial \eta^2} + d \frac{\partial u}{\partial \eta} + e \frac{\partial u}{\partial \xi} \right) = f$$

$$\frac{-1}{J^2} (a u_{\xi\xi} - 2b u_{\xi\eta} + c u_{\eta\eta} + d u_{\eta} + e u_{\xi}) = f$$

- Here the a, b, c, d , and e are associated only with the mathematics of the mapping.
- It is also worth noting that $a(\xi, \eta), b(\xi, \eta), c(\xi, \eta), d(\xi, \eta)$, and $e(\xi, \eta)$ are functions of the mapping coordinates – meaning that each point in the computational domain may take a different of these variables.

Poisson Equation Mapping and General 2D Domains

- Where, a, b, c, d , and e are intermediate variables that depend ONLY on the coordinates/mapping as:

$$a = x_\eta^2 + y_\eta^2 \quad e = \frac{x_\eta \beta - y_\eta \alpha}{J} \quad \alpha = a x_{\xi\xi} - 2b x_{\xi\eta} + c x_{\eta\eta} \quad (2)$$

$$b = x_\eta x_\xi + y_\eta y_\xi \quad d = \frac{y_\xi \alpha - x_\xi \beta}{J} \quad \beta = a y_{\xi\xi} - 2b y_{\xi\eta} + c y_{\eta\eta} \quad (3)$$

$$c = x_\xi^2 + y_\xi^2 \quad J = x_\xi y_\eta - x_\eta y_\xi \quad (4)$$

Equation Discretization Using finite differences at each internal node

$$\frac{-1}{J^2} (au_{\xi\xi} - 2bu_{\xi\eta} + cu_{\eta\eta} + du_{\eta} + eu_{\xi}) = f \quad (5)$$

Let's give this a try:

$$\begin{aligned} f &= \frac{-a}{J^2} \left(\frac{u_{i-1,j} - 2u_{i,j} + u_{i+1,j}}{\Delta\xi^2} \right) \\ &+ \frac{2b}{J^2} \left(\frac{u_{i+1,j+1} - u_{i+1,j-1} - u_{i-1,j+1} + u_{i-1,j-1}}{4\Delta\eta\Delta\xi} \right) \\ &+ \frac{-c}{J^2} \left(\frac{u_{i,j-1} - 2u_{i,j} + u_{i,j+1}}{\Delta\eta^2} \right) \\ &+ \frac{-d}{J^2} \left(\frac{u_{i,j+1} - u_{i,j-1}}{2\Delta\eta} \right) \\ &+ \frac{-e}{J^2} \left(\frac{u_{i+1,j} - u_{i-1,j}}{2\Delta\xi} \right) \\ &= f \end{aligned}$$

Equation Discretization: Determining the A-Matrix Coefficients

- The previous equation has unknowns “everywhere” – what we really want to do is collect the coefficients for each unknown into a single coefficient.
- The form of this equation is:

$$\begin{aligned} AA * u_{i+1,j+1} + BB * u_{i+1,j} + CC * u_{i+1,j-1} &+ \\ DD * u_{i,j+1} + EE * u_{i,j} + FF * u_{i,j-1} &+ \\ GG * u_{i-1,j+1} + HH * u_{i-1,j} + KK * u_{i-1,j-1} &= F_{i,j} \end{aligned}$$

- What are, $AA, BB, CC, DD, EE, FF, GG, HH, KK$?

Collecting the FD and Mapping coefficients – stencil center node EE relates to $u_{i,j}$

- According to our definition, the coefficient relates to unknowns as:
 - Coefficient $EE \rightarrow u_{i,j}$;
- Equation which determines the coefficients:

$$\begin{aligned} f = & \frac{-a}{J^2} \left(\frac{u_{i-1,j} - 2u_{i,j} + u_{i+1,j}}{\Delta\xi^2} \right) \\ & + \frac{2b}{J^2} \left(\frac{u_{i+1,j+1} - u_{i+1,j-1} - u_{i-1,j+1} + u_{i-1,j-1}}{4\Delta\eta\Delta\xi} \right) \\ & + \frac{-c}{J^2} \left(\frac{u_{i,j-1} - 2u_{i,j} + u_{i,j+1}}{\Delta\eta^2} \right) \\ & + \frac{-d}{J^2} \left(\frac{u_{i,j+1} - u_{i,j-1}}{2\Delta\eta} \right) + \frac{-e}{J^2} \left(\frac{u_{i+1,j} - u_{i-1,j}}{2\Delta\xi} \right) = f \end{aligned}$$

- So, the EE-coefficient is:

$$EE = \frac{2}{J^2} \left(\frac{a}{\Delta\xi^2} + \frac{c}{\Delta\eta^2} \right)$$

Collecting the FD and Mapping coefficients for unknowns

– $u_{i\pm 1,j\pm 1} \rightarrow AA, CC, GG, KK$

- According to our definition, the coefficients relate to unknowns as:

- Coefficient $AA \rightarrow u_{i+1,j+1}$;
- Coefficient $CC \rightarrow u_{i+1,j-1}$;
- Coefficient $GG \rightarrow u_{i-1,j+1}$
- Coefficient $KK \rightarrow u_{i-1,j-1}$;

- Equation which determines the coefficients:

$$\begin{aligned} f &= \frac{-a}{J^2} \left(\frac{u_{i-1,j} - 2u_{i,j} + u_{i+1,j}}{\Delta \xi^2} \right) \\ &+ \frac{2b}{J^2} \left(\frac{u_{i+1,j+1} - u_{i+1,j-1} - u_{i-1,j+1} + u_{i-1,j-1}}{4\Delta \eta \Delta \xi} \right) \\ &+ \frac{-c}{J^2} \left(\frac{u_{i,j-1} - 2u_{i,j} + u_{i,j+1}}{\Delta \eta^2} \right) \\ &+ \frac{-d}{J^2} \left(\frac{u_{i,j+1} - u_{i,j-1}}{2\Delta \eta} \right) + \frac{-e}{J^2} \left(\frac{u_{i+1,j} - u_{i-1,j}}{2\Delta \xi} \right) = f \end{aligned}$$

- So, the AA, CC, GG, and KK -coefficients are:

$$AA = +\frac{2b}{J^2} \frac{1}{4\Delta \eta \Delta \xi}, \quad CC = -\frac{2b}{J^2} \frac{1}{4\Delta \eta \Delta \xi}, \quad GG = -\frac{2b}{J^2} \frac{1}{4\Delta \eta \Delta \xi}, \quad KK = +\frac{2b}{J^2} \frac{1}{4\Delta \eta \Delta \xi}$$

Collecting the FD and Mapping coefficients for each – for BB, DD, FF, HH

- According to our definition, the coefficients relate to unknowns as:

- Coefficient $BB \rightarrow u_{i+1,j}$; Coefficient $DD \rightarrow u_{i,j+1}$;
- Coefficient $FF \rightarrow u_{i,j-1}$ Coefficient $HH \rightarrow u_{i-1,j}$;

- Equation which determines the coefficients:

$$\begin{aligned} f &= \frac{-a}{J^2} \left(\frac{u_{i-1,j} - 2u_{i,j} + u_{i+1,j}}{\Delta\xi^2} \right) \\ &+ \frac{2b}{J^2} \left(\frac{u_{i+1,j+1} - u_{i+1,j-1} - u_{i-1,j+1} + u_{i-1,j-1}}{4\Delta\eta\Delta\xi} \right) \\ &+ \frac{-c}{J^2} \left(\frac{u_{i,j-1} - 2u_{i,j} + u_{i,j+1}}{\Delta\eta^2} \right) \\ &+ \frac{-d}{J^2} \left(\frac{u_{i,j+1} - u_{i,j-1}}{2\Delta\eta} \right) + \frac{-e}{J^2} \left(\frac{u_{i+1,j} - u_{i-1,j}}{2\Delta\xi} \right) = f \end{aligned}$$

- So, the BB, DD, FF , and HH -coefficients are:

$$\begin{aligned} BB &= \frac{-1}{J^2} \left(\frac{a}{\Delta\xi^2} + \frac{e}{2\Delta\xi} \right), \quad DD = \frac{-1}{J^2} \left(\frac{c}{\Delta\eta^2} + \frac{d}{2\Delta\eta} \right) \\ FF &= \frac{-1}{J^2} \left(\frac{c}{\Delta\eta^2} - \frac{d}{2\Delta\eta} \right), \quad HH = \frac{-1}{J^2} \left(\frac{a}{\Delta\xi^2} - \frac{e}{2\Delta\xi} \right) \end{aligned}$$

Determining A-Matrix Coefficients – Organizing the equation

- Now we can see:

$$\begin{aligned}F_{i,j} &= AA * u_{i+1,j+1} + BB * u_{i+1,j} + CC * u_{i+1,j-1} \\&+ DD * u_{i,j+1} + EE * u_{i,j} + FF * u_{i,j-1} \\&+ GG * u_{i-1,j+1} + HH * u_{i-1,j} + KK * u_{i-1,j-1}\end{aligned}$$

- With, $AA, BB, CC, DD, EE, FF, GG, HH, KK$:

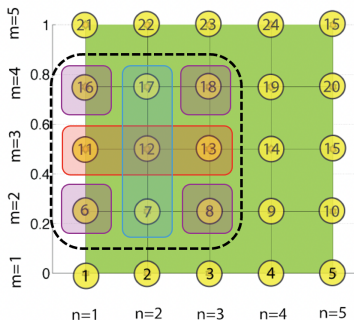
$$AA = +\frac{2b}{J^2} \frac{1}{4\Delta\eta\Delta\xi}, \quad BB = \frac{-1}{J^2} \left(\frac{a}{\Delta\xi^2} + \frac{e}{2\Delta\xi} \right), \quad CC = -\frac{2b}{J^2} \frac{1}{4\Delta\eta\Delta\xi},$$

$$DD = \frac{-1}{J^2} \left(\frac{c}{\Delta\eta^2} + \frac{d}{2\Delta\eta} \right) \quad EE = \frac{2}{J^2} \left(\frac{a}{\Delta\xi^2} + \frac{c}{\Delta\eta^2} \right), \quad FF = \frac{-1}{J^2} \left(\frac{c}{\Delta\eta^2} - \frac{d}{2\Delta\eta} \right),$$

$$GG = -\frac{2b}{J^2} \frac{1}{4\Delta\eta\Delta\xi}, \quad HH = \frac{-1}{J^2} \left(\frac{a}{\Delta\xi^2} - \frac{e}{2\Delta\xi} \right), \quad KK = +\frac{2b}{J^2} \frac{1}{4\Delta\eta\Delta\xi}$$

Determining A-Matrix Coefficients – Organizing the equation

- Considering node 12 ($n, m = (2, 3) \rightarrow \text{Node12}$):



- The discrete equation applied at node 12 is (in global node numbering):

$$f_{12} = AA * u_{18} + BB * u_{13} + CC * u_8 + DD * u_{17} + EE * u_{12} + FF * u_7 \\ + GG * u_{16} + HH * u_{11} + KK * u_6$$

- The result is a “9-node” difference equation stencil in row 12 of the A-matrix.

A-Matrix – Representing the Linear System of Equations (one equation per node)

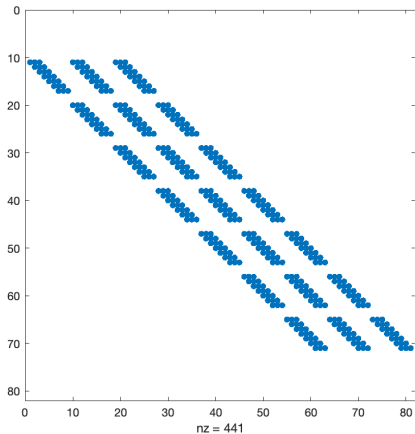
- For node 12, we can write the equation as a single row of a Matrix system:

$$\begin{aligned}
 f_{12} &= AA * u_{18} + BB * u_{13} + CC * u_8 \\
 &+ DD * u_{17} + EE * u_{12} + FF * u_7 \\
 &+ GG * u_{16} + HH * u_{11} + KK * u_6 = f_{12}
 \end{aligned}$$

$$\begin{bmatrix}
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 : & : & : & : & : & : & : & : & : & : & : & : & : \\
 : & : & : & : & : & : & : & : & : & : & : & : & : \\
 KK & FF & CC & \dots 0 \dots & HH & EE & BB & \dots 0 \dots & GG & DD & AA & : & : \\
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 \end{bmatrix}
 \begin{pmatrix}
 u_6 \\
 u_7 \\
 u_8 \\
 : \\
 u_{11} \\
 u_{12} \\
 u_{13} \\
 : \\
 u_{16} \\
 u_{17} \\
 u_{18}
 \end{pmatrix}
 =
 \begin{pmatrix}
 : \\
 : \\
 : \\
 : \\
 : \\
 f_{12} \\
 : \\
 : \\
 : \\
 : \\
 :
 \end{pmatrix}$$

A-Matrix – Representing the Linear System of Equations (one equation per node)

- After the coefficients for each node's discrete difference equation are placed in the matrix, the matrix structure is 3-bands (diagonal) of 3 entries:



- These bands of values connect the solution at one node to every other node.

The Finite Difference Stencil for this Problem

- The discrete equation representing the PDE at a node in the domain will form a “stencil” with the grid $n, m = i, j$ indices:

$u_{i-1,j+1} : GG : Northwest$	$u_{i,j+1} : DD : North$	$u_{i+1,j+1} : AA : Northeast$
$u_{i-1,j} : HH : West$	$u_{i,j} : EE : Center$	$u_{i+1,j} : BB : East$
$u_{i-1,j-1} : KK : Southwest$	$u_{i,j-1} : FF : South$	$u_{i+1,j-1} : CC : Southeast$

- This corresponds to the following equation coefficients/matrix entries for these locations:

$-\frac{2b}{J^2} \frac{1}{4\Delta\eta\Delta\xi}$	$\frac{-1}{J^2} \left(\frac{c}{\Delta\eta^2} + \frac{d}{2\Delta\eta} \right)$	$+\frac{2b}{J^2} \frac{1}{4\Delta\eta\Delta\xi}$
$\frac{-1}{J^2} \left(\frac{a}{\Delta\xi^2} - \frac{e}{2\Delta\xi} \right)$	$\frac{2}{J^2} \left(\frac{a}{\Delta\xi^2} + \frac{c}{\Delta\eta^2} \right)$	$\frac{-1}{J^2} \left(\frac{a}{\Delta\xi^2} + \frac{e}{2\Delta\xi} \right)$
$+\frac{2b}{J^2} \frac{1}{4\Delta\eta\Delta\xi}$	$\frac{-1}{J^2} \left(\frac{c}{\Delta\eta^2} - \frac{d}{2\Delta\eta} \right)$	$-\frac{2b}{J^2} \frac{1}{4\Delta\eta\Delta\xi}$

- These coefficients (entries in the A-matrix), need to to be inserted appropriately for each **internal node**.
 - Row of [A] corresponding to equation/node number.
 - Column of [A] corresponding to node number associated with stencil location (N, NE, E, SE, W, etc.).

Example: Implementing the A-matrix, Steps with an Example

- After we complete the mapping/geometry definition, we want to actually build the A-Matrix.
- We can do this by setting up a pair of for-loops (nested for loops) to cycle over the internal nodes.
- Inside that nested for loop, we will calculate the entries into the A-matrix

```
for(m = 2:(M-1))  
for(n = 2:(N-1))  
% A matrix assembly will go here  
end  
end
```

Example: Implementing the A-matrix, Steps - mapping derivatives

- Consider the following mapping:

$$\begin{aligned}x(\xi, \eta) &= R_x \xi \eta + S_x \xi + T_x \eta + U_x \\y(\xi, \eta) &= R_y \xi \eta + S_y \xi + T_y \eta + U_y\end{aligned}$$

- Calculate the derivatives associated with the specific mapping (some examples below):

$$\begin{aligned}x_\xi &= \frac{\partial x}{\partial \xi} = R_x \eta + S_x, & x_\eta &= \frac{\partial x}{\partial \eta} = R_x \xi + T_x \\y_\xi &= \frac{\partial y}{\partial \xi} = R_y \eta + S_y, & y_\eta &= \frac{\partial y}{\partial \eta} = R_y \xi + T_y\end{aligned}$$

- If you have a numerical mapping, then, finite differences will be used:

$$\begin{aligned}x_\xi &= \frac{\partial x}{\partial \xi} = \frac{x_{i+1,j} - x_{i-1,j}}{2\Delta\xi}, & x_\eta &= \frac{\partial x}{\partial \eta} = \frac{x_{i,j+1} - x_{i,j-1}}{2\Delta\eta} \\y_\xi &= \frac{\partial y}{\partial \xi} = \frac{y_{i+1,j} - y_{i-1,j}}{2\Delta\xi}, & y_\eta &= \frac{\partial y}{\partial \eta} = \frac{y_{i,j+1} - y_{i,j-1}}{2\Delta\eta}\end{aligned}$$

Example: Mapping coefficients

- Once you have all the derivatives for the mapping calculated at each internal node (note; you can do this as you populate $[A]$, “on-the-fly”), determine the equation coefficients:

$$a = x_\eta^2 + y_\eta^2 \quad e = \frac{x_\eta \beta - y_\eta \alpha}{J} \quad \alpha = a x_{\xi\xi} - 2b x_{\xi\eta} + c x_{\eta\eta} \quad (6)$$

$$b = x_\eta x_\xi + y_\eta y_\xi \quad d = \frac{y_\xi \alpha - x_\xi \beta}{J} \quad \beta = a y_{\xi\xi} - 2b y_{\xi\eta} + c y_{\eta\eta} \quad (7)$$

$$c = x_\xi^2 + y_\xi^2 \quad J = x_\xi y_\eta - x_\eta y_\xi \quad (8)$$

Example: Populating the entries of the A-matrix

- For each node, calculate (AA, BB, CC,...,KK) and populate these entries of the matrix [A] (on the appropriate row).
 - Current row $rowNum = NN(n,m)$
 - Some example entries:
 - $A(rowNum, NN(n,m)) = EE;$
 - $A(rowNum, NN(n+1,m)) = BB;$
 - $A(rowNum, NN(n,m+1)) = DD;$
 - etc.
 - Populate the RHS vector, f , with the appropriate value for this node, $f(rowNum, 1) = val.$
- At this point you should have an A-matrix populated with the equations for the internal nodes.

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Boundary Conditions

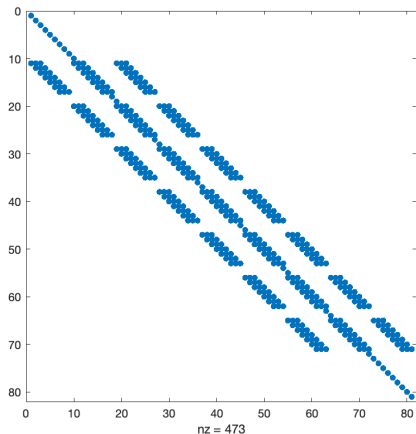
- We will implement simple **Dirichlet Boundary Conditions**:

$$1 * u_{i,j} = 0$$

- Recall from 2D Finite Differences, that we can insert this as an equation in the A-matrix by;
 - Setting the whole row corresponding to the node to zero.
 - Assigning a value of 1 to the diagonal of that row.
 - Assigning the $f_{i,j}$ value to the appropriate value
 - In our example membrane problem, we'll set the boundary to zero deflection.
- See code from the 2-D Finite Difference solver we made on regular grids for code implementation details.

Boundary Conditions

- Once we implement **Dirichlet Boundary Conditions** the matrix will take the form:



- As we can see, now each row has an equation represented (entries). We should be able to solve now assuming none of the equations/columns of $[A]$ are linearly dependent.

Main Steps in the Development of a 2D Poisson solver with Domain Mapping

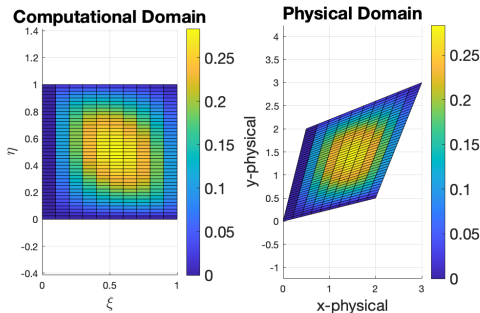
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Linear System Solution and Post-processing

- The process for solving is identical to that used before for a linear system of equations in matrix form:

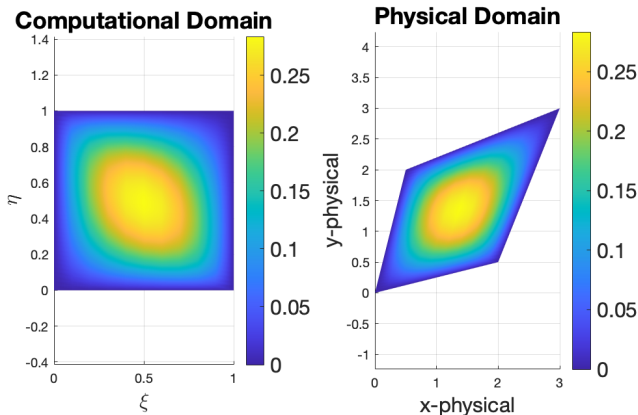
$$[A][u] = [f]$$

- The post processing is also identical to that before.



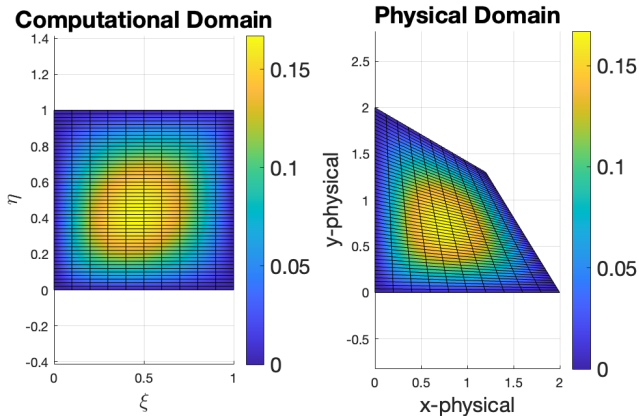
Linear System Solution and Post-processing

- Here we smooth the plot by having MATLAB interpolate the resulting colormap.



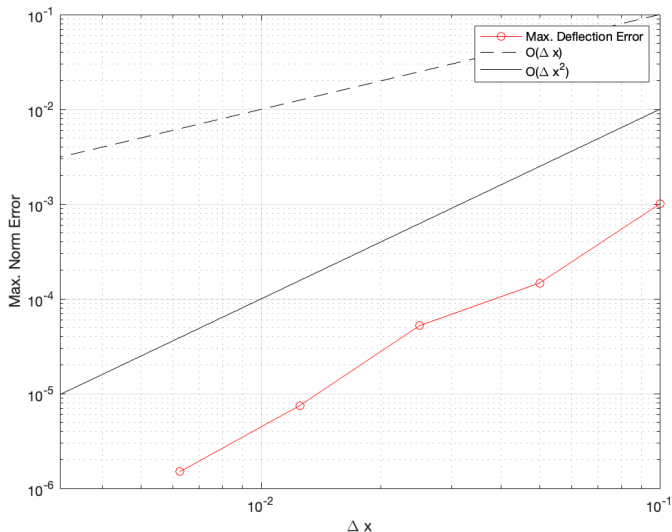
Linear System Solution and Post-processing

- Notice here how we are able to refine in one direction more than the other and still maintain a reasonable solution.



Linear System Solution and Post-processing

- We should also have second order accurate error convergence.



What did we learn?

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